

Macroeconomics II, Lecture XII: The Buffer-Stock Savings Model

Erik Öberg

- Last lecture: theoretical implications of uninsurable income risk for consumption-savings dynamics
 - ① With incomplete markets, ex-ante homogeneous households will be **ex-post heterogeneous** in terms of $\{C, A, Y\} \Rightarrow$ no aggregation into representative agent
 - ② With incomplete markets, consumption dynamics influenced by a **precautionary savings** motive
- Today: using incomplete-markets consumption-savings theory for quantitative empirical analysis
 - ▶ Research program pioneered by Deaton, Zeldes, and Carroll
 - ▶ Builds on foundational work by Friedman, Modigliani, Hall and others

- ❶ Formulating and solving the canonical buffer-stock savings model
 - ▶ Recursive formulation
 - ▶ Solution algorithm
 - ▶ Calibration
- ❷ Consumption-savings dynamics in the buffer-stock savings model
 - ▶ Consumption-savings dynamics without income risk
 - ▶ Consumption-savings dynamics with income risk
- ❸ Three influential applications
 - ❶ Gourinchas-Parker (Econometrica 2002): Consumption over the Life Cycle
 - ❷ Blundell-Pistaferri-Preston (AER 2008): Consumption Inequality and Partial Insurance
 - ❸ Kaplan-Violante (Econometrica 2014): A Model of the Consumption Response to Fiscal Stimulus Payments

The canonical buffer-stock savings model

- The buffer-stock savings model = income-fluctuations problem with persistent shocks
 - ▶ Key papers: Zeldes (JPE 1989; QJE 1989), Deaton (Econometrica 1991, book 1992), Carroll (QJE 1997)
- Household problem:

$$\begin{aligned} \max_{C_t, A_{t+1}} \quad & E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma} \\ \text{s.t.} \quad & C_t + A_{t+1} = Y_t + RA_t \\ & Y_t = P_t e^{\epsilon_t} \\ & P_t = P_{t-1} G e^{\nu_t} \\ & A_{t+1} \geq -\underline{A} \\ & C_t \geq 0 \end{aligned}$$

The canonical buffer-stock savings model: terminology

- Terminology:
 - ▶ P_t — permanent income
 - ▶ ν_t — permanent income shocks
 - ▶ ϵ_t — transitory income shocks
- We assume that ϵ_t, ν_t are known at the time of choosing C_t, A_{t+1}

- G is a constant for simplicity
 - ▶ It can be any deterministic function
 - ▶ In many applications, it is a function of age, education, ability etc.
- Why a permanent-transitory formulation of the income process?
 - ▶ In the data, we see that some changes to residualized income are very persistent, whereas others are very short-lived
 - ▶ The permanent-transitory formulation provides a parsimonious parameterization of the household income process that captures these features
- CRRA utility: a reasonable baseline with large tractability gains
 - ▶ Allows us to normalize the problem with respect to permanent income

Recursive formulation

- To solve the problem and investigate its properties, we need to recast it in recursive form
- Introduce cash on hand $M_t = Y_t + RA_t$
- Recursive formulation

$$\begin{aligned} V(\quad) &= \max_{C, A'} U(C) + \beta EV(\quad) \\ \text{s.t. } &A' = M - C \\ &M' = P'e^{\epsilon'} + RA' \\ &P' = GPe^{\nu'} \\ &A' \geq -\underline{A} \\ &C \geq 0 \end{aligned}$$

- Which are the state variables?
 - What information, known at time t , is useful for the household in choosing C, A' and computing $U(C) + \beta EV'()$?

Recursive formulation

- To solve the problem and investigate its properties, we need to recast it in recursive form
- Introduce cash on hand $M_t = Y_t + RA_t$
- Recursive formulation

$$\begin{aligned} V(M, P) &= \max_{C, A'} U(C) + \beta EV(M', P') \\ \text{s.t. } &A' = M - C \\ &M' = P' e^{\epsilon'} + RA' \\ &P' = GPe^{\nu'} \\ &A' \geq -\underline{A} \\ &C \geq 0 \end{aligned}$$

- Which are the state variables?
 - ▶ What information, known at time t , is useful for the household in choosing C, A' and computing $U(C) + \beta EV'()$?

Normalization with respect to permanent income

- It appears that the state variables are cash on hand M and permanent income P
- With $U(C) = \frac{C^{1-\sigma}}{1-\sigma}$ being CRRA, permanent income is actually not a state variable
- Define
 - ▶ $m = \frac{M}{P}$
 - ▶ $c = \frac{C}{P}$
 - ▶ $a' = \frac{A'}{P}$
 - ▶ $\underline{a} = \frac{A}{P}$
 - ▶ $v(M, P) = \frac{V(M, P)}{P^{1-\sigma}}$

- Household problem

$$\begin{aligned} V(M, P) &= \max_{C, A'} \frac{C^{1-\sigma}}{1-\sigma} + \beta EV(M', P') \\ \text{s.t. } &A' = M - C \\ &M' = P' e^{\epsilon'} + RA' \\ &P' = GPe^{\nu'} \\ &A' \geq -\underline{A} \\ &C \geq 0 \end{aligned}$$

Normalization with respect to permanent income II

- Using our definitions

$$\begin{aligned} P^{1-\sigma} v(M, P) &= \max_{c, a'} \frac{(cP)^{1-\sigma}}{1-\sigma} + \beta E P'^{1-\sigma} v(M', P') \\ \text{s.t. } a' P &= mP - cP \\ m' P' &= P' e^{\epsilon'} + R a' P \\ P' &= G P e^{\nu'} \\ a' P &\geq -\underline{a} P \\ cP &\geq 0 \end{aligned}$$

- Dividing through

$$\begin{aligned}v(M, P) &= \max_{c, a'} \frac{c^{1-\sigma}}{1-\sigma} + \beta E \left(\frac{P'}{P} \right)^{1-\sigma} v(M', P') \\ \text{s.t. } &a' = m - c \\ &m' = e^{\epsilon'} + Ra' \frac{P}{P'} \\ &\frac{P'}{P} = Ge^{\nu'} \\ &a' \geq -\underline{a} \\ &c \geq 0\end{aligned}$$

- Substituting the law-of-motion for permanent income:

$$\begin{aligned}v(M, P) &= \max_{c, a'} \frac{c^{1-\sigma}}{1-\sigma} + \beta E \left(Ge^{\nu'} \right)^{1-\sigma} v(M', P') \\ \text{s.t. } &a' = m - c \\ &m' = e^{\epsilon'} + \frac{Ra'}{Ge^{\nu'}} \\ &a' \geq -\underline{a} \\ &c \geq 0\end{aligned}$$

- Normalized household problem:

$$\begin{aligned}v(M, P) &= \max_{c, a'} \frac{c^{1-\sigma}}{1-\sigma} + \beta E_t \left(Ge^{\nu'} \right)^{1-\sigma} v(M', P') \\ \text{s.t. } &a' = m - c \\ &m' = e^{\epsilon'} + \frac{Ra'}{Ge^{\nu'}} \\ &a' \geq -\underline{a} \\ &c \geq 0\end{aligned}$$

- Which are the state variables?

- Normalized household problem:

$$\begin{aligned}v(m) &= \max_{c, a'} \frac{c^{1-\sigma}}{1-\sigma} + \beta E_t \left(Ge^{\nu'} \right)^{1-\sigma} v(m') \\ \text{s.t. } &a' = m - c \\ &m' = e^{\epsilon'} + \frac{Ra'}{Ge^{\nu'}} \\ &a' \geq -\underline{a} \\ &c \geq 0\end{aligned}$$

- Which are the state variables?

What did we just learn?

- The sufficient state variable is $m = \frac{M}{P}$
- Economics:
 - ▶ Decision functions in normalized problem: $c = c(m)$, $a' = a'(m)$
 - ▶ Unnormalized decision functions $C = Pc(m)$, $A' = Pa'(m)$
 - ▶ Implication: a permanent-income rich household will consume the same fraction of an increase in normalized cash-on-hand as a permanent-income poor household
- Computations:
 - ▶ We have reduced a two-dimensional functional equation to a one-dimensional functional equation
 - ▶ Big computational gain of dimension reduction when solving the problem numerically
- Key assumptions: CRRA utility (or, more generally, a power-function utility) and linear constraints

Properties of solution

- Solution given by a consumption function $c(m)$ and a savings function $a'(m) = m - c(m)$

- As usual, an interior solution $c(m)$ must satisfy the Euler equation

$$c^{-\sigma} = \beta RE \left(\left(Ge^{\nu'} c' \right)^{-\sigma} \right)$$

- Else, the credit constraint is binding and $c(m) = m + \underline{a}$
- Given some parametric restrictions, the Bellman equation defines a contraction mapping, and we can solve the equation using value function iteration

Value function iteration

- Construct a discrete grid $\hat{m} = \{\hat{m}_1, \hat{m}_2, \dots, \hat{m}_N\}$
- Guess a discrete value function $\hat{v}^0 = \{\hat{v}_1^0, \hat{v}_2^0, \dots, \hat{v}_N^0\}$
- For each $\hat{m}_k$, $k = 1, \dots, N$, solve the decision problem

$$\begin{aligned} \max_{c, a'} \quad & \frac{c^{1-\sigma}}{1-\sigma} + \beta \sum_i \sum_j \pi(\eta_i) \pi(\epsilon_j) \left(Ge^{\nu_i'} \right)^{1-\sigma} \hat{v}^0 \left(e^{\epsilon_j} + \frac{Ra'}{Ge^{\nu_i}} \right) \\ \text{s.t.} \quad & a' = \hat{m}_k - c \\ & a' \geq -\underline{a} \\ & c \geq 0 \end{aligned}$$

using a standard numerical maximization method (e.g. Newton-Raphson)

Value function iteration: updating the guess

- If $\hat{m}_k < e^{\epsilon_j} + \frac{Ra'_i}{Ge^{\nu_i}} < \hat{m}_{k+1}$ for some $k < N \Rightarrow$ interpolate v_k^0, v_{k+1}^0 to compute $\hat{v}^0\left(e^{\epsilon_j} + \frac{Ra'_i}{Ge^{\nu_i}}\right)$
- The maximum of the objective function is your new guess $\hat{v}_k^1$
- Repeat until convergence

- Parameters: $\beta, \sigma, G, R, \underline{a}$ and the distribution of ϵ', ν'
- How to select parameter values?
- G and R can be directly observed in the data
- The distribution of ϵ', ν' can be estimated using panel data on household income
 - ▶ Survey data, e.g., CPS, PSID (US)
 - ▶ Register data, e.g., Swedish income registers
- Calibration of $\beta, \sigma, \underline{a}$ is typically done by method of moments using data on household income and consumption
 - ▶ Survey data, e.g., PSID-CEX (US)
 - ▶ Register data, e.g., Swedish income and wealth registers

Income process estimation

- In logs ($\tilde{x} = \log X$), our process is

$$\begin{aligned} Y_t &= P_t e^{\epsilon_t}, & y_t &= p_t + \epsilon_t \\ P_t &= P_{t-1} G e^{\nu_t}, & p_t &= p_{t-1} + g + \nu_t \end{aligned}$$

which gives us

$$\begin{aligned} \Delta y_t &= g + \nu_t + \epsilon_t - \epsilon_{t-1} \\ \Delta y_{t-1} &= g + \nu_{t-1} + \epsilon_{t-1} - \epsilon_{t-2} \end{aligned}$$

which means that

$$\begin{aligned} \text{Var}(\Delta y_t) &= \text{Var}(\nu_t) + 2\text{Var}(\epsilon_t) \\ \text{Cov}(\Delta y_t, \Delta y_{t-1}) &= -\text{Var}(\epsilon_t) \end{aligned}$$

- Typically, we apply this estimator to income growth residuals in household panel data
- This method can be generalized in several dimensions, and recent administrative data have taught us a lot:
 - ▶ Guvenen-Karahan-Ozkan-Song (Econometrica 2021): income growth residuals exhibit fat tails and significant skewness (normality is a bad assumption)
 - ▶ Carter Braxton-Herkenhoff-Rothbaum-Schmidt (AER 2025): permanent earnings risk has evolved differently across the skill distribution over time

Consumption-savings dynamics in the buffer-stock savings model

- Now we know how to solve and parameterize our model
- Next step: look at how households behave in the model
- Recall:
 - ▶ state variable: $m = \frac{M}{P} = \frac{Y+RA}{P}$
 - ▶ law of motion: $m' = e^{\epsilon'} + \frac{Ra'(m)}{Ge^{\nu'}}$
 - ▶ decisions: $c = c(m)$ and $a' = a'(m) = m - c(m)$
 - ▶ unnormalized decisions: $C = c(m)P$ and $A' = a'(m)P$
- Defining features of the model: uninsurable income risk and potentially binding credit constraint
- To understand the implications of these features, we first remind ourselves about consumption-savings dynamics in a frictionless perfect-foresight model

Perfect-foresight model

- Set $\epsilon = \nu = 0$ and $\underline{a} = \underline{a}^{nbl}$, where $\underline{a}^{nbl}$ is the natural borrowing limit
- Perfect-foresight Bellman equation:

$$\begin{aligned} v(m) &= \max_{c, a'} \frac{c^{1-\sigma}}{1-\sigma} + \beta G^{1-\sigma} v(m') \\ \text{s.t. } a' &= m - c \\ m' &= 1 + \frac{Ra'}{G} \\ a' &\geq -\underline{a}^{nbl} \\ c &\geq 0 \end{aligned}$$

- As we will iterate on the budget constraint, I now reintroduce time subscripts
- As $\underline{a} = \underline{a}^{nbl}$, we will always have an interior solution
- Interior solution satisfies

$$c_t^{-\sigma} = \beta R (G c_{t+1})^{-\sigma}$$

- For simplicity, set σ so that $(\beta R)^{-\frac{1}{\sigma}} G = 1 \Rightarrow$

$$c_t = c_{t+1}$$

- To solve for $c = c(m)$, use Euler equation and iterate on the budget constraint and the law-of-motion for m :

$$a_{t+1} = m_t - c_t$$

$$m_{t+1} = 1 + \frac{Ra_{t+1}}{G}$$

Perfect-foresight model: solution

- Substituting and iterating forward gives us

$$\begin{aligned}m_{t+1} &= 1 + \frac{R}{G}(m_t - c_t) \\ \Rightarrow m_t &= c_t - \frac{G}{R} + \frac{G}{R}m_{t+1} \\ \Rightarrow m_t &= c_t - \frac{G}{R} + \frac{G}{R}\left(c_{t+1} - \frac{G}{R} + \frac{G}{R}m_{t+2}\right) \\ &\dots \\ \Rightarrow m_t &= \sum_{k=0}^{\infty} \left(\frac{G}{R}\right)^k c_{t+k} - \sum_{k=1}^{\infty} \left(\frac{G}{R}\right)^k + \lim_{T \rightarrow \infty} \left(\frac{G}{R}\right)^T m_{t+T} \\ \Rightarrow m_t &= c_t \frac{1}{1 - \frac{G}{R}} - \frac{G}{R} \frac{1}{1 - \frac{G}{R}}\end{aligned}$$

where we have used that $c_{t+k} = c_t$ for all k and the transversality condition

$$\lim_{T \rightarrow \infty} \left(\frac{G}{R}\right)^T m_{t+T} = 0$$

- Therefore, we get

$$c_t = \left(1 - \underbrace{\frac{G}{R}}_{\text{eff. disc. rate}} \right) \left(\underbrace{m_t}_{\text{c.o.h.}} + \underbrace{\frac{G}{R} \frac{1}{1 - \frac{G}{R}}}_{\text{NPV future norm. inc.}} \right)$$

- $\Rightarrow$ households behave according to the permanent-income hypothesis (PIH)

Perfect-foresight model: MPC out of transitory income shocks

- Consider a marginal change in M , holding P constant:
 - ▶ I.e., consider a transitory shock to current income

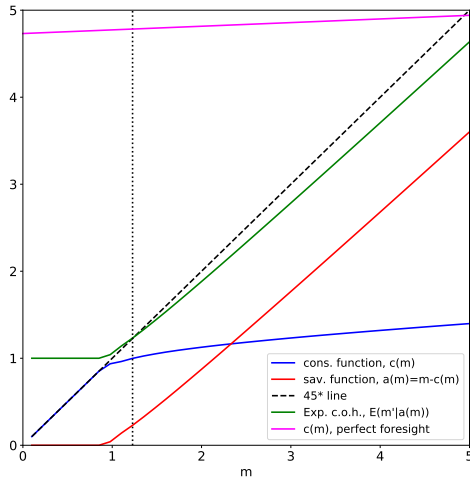
$$\begin{aligned}\frac{\partial C}{\partial M} &= \frac{\partial (c(m)P)}{\partial (mP)} \\ &= \frac{P \partial (c(m))}{P \partial m} \\ &= \frac{\partial c(m)}{\partial m}\end{aligned}$$

which gives us

$$\frac{\partial C}{\partial M} = \left(1 - \frac{G}{R}\right)$$

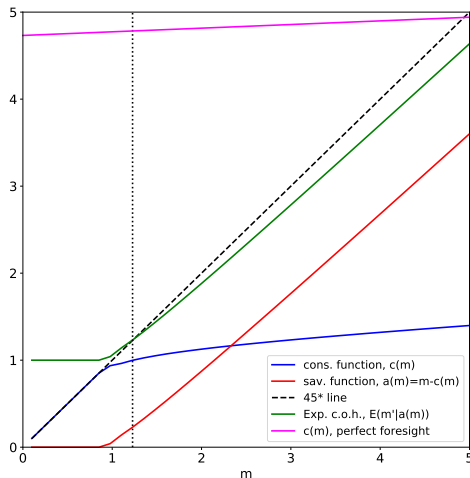
- Insights:
 - ▶ Marginal propensity to consume (MPC) out of current income is constant
 - ▶ For reasonable parameters: MPC out of current income is very small

Consumption dynamics in the buffer-stock savings model



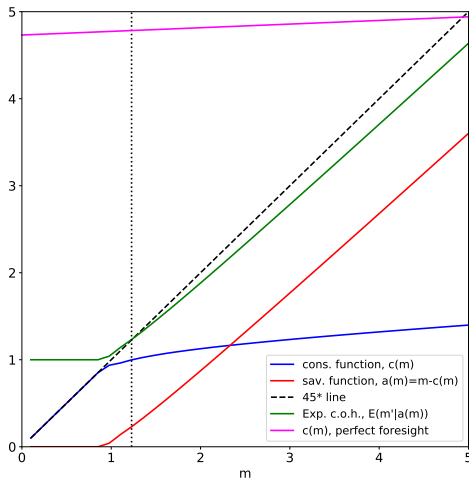
- Parameter values: $\beta = 0.95$, $\sigma = 1.5$, $R = 1.04$, $G = 1.03$, $\underline{a} = 0$, with $\sigma_\epsilon, \sigma_\nu$ taken from Gourinchas-Parker (Econometrica, 2002)
- Code available by email

Consumption dynamics in the buffer-stock savings model



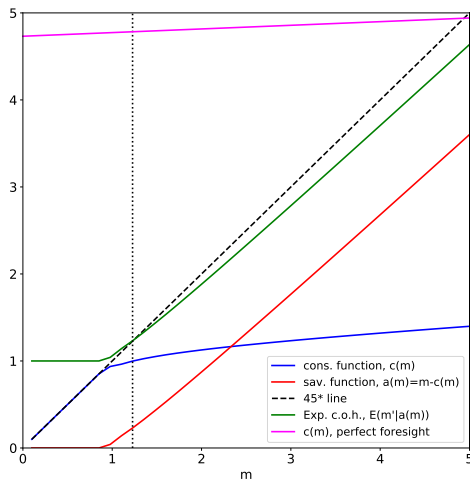
- Result 1: with income risk, households consume less than they would without income risk
 - ▶ with income risk, assets have insurance value
 - ▶ without income risk, households run down current assets quickly due to impatience

Consumption dynamics in the buffer-stock savings model



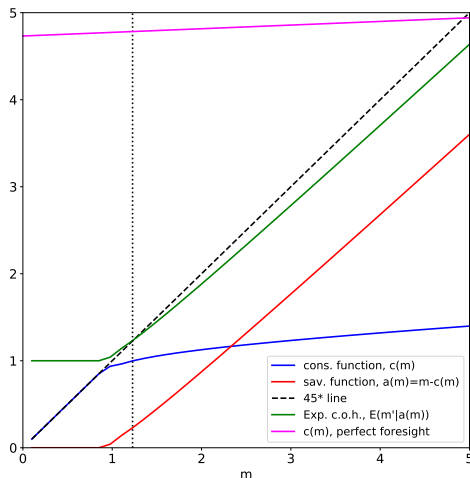
- Result 2: with income risk, the consumption function is concave
 - ▶ Equivalently: with income risk, the MPC out of current income is larger and decreasing
 - ▶ As the household approaches the constraint, the MPC out of current income grows to 1

Consumption dynamics in the buffer-stock savings model



- With a lot of assets, households behave as if there were no income risk, and $MPC \rightarrow \left(1 - \frac{G}{R}\right)$

Consumption dynamics in the buffer-stock savings model



- Result 3: because of concavity, there is a target buffer stock of assets
 - ▶ To insure themselves, households seek to hold a certain amount of money relative to their permanent income
 - ▶ With a lot of assets, households dissave

- Now we understand the basic properties of the buffer-stock savings model
- Move on to analyze how to use the model to interpret the data
- Early literature (Zeldes, Deaton, Carroll, and others) focused on testing the buffer-stock model against the PIH model (this description does not fit all papers)
 - ▶ Perhaps ex post unsurprisingly, PIH was rejected in several dimensions
 - ▶ In the late 1980s/early 1990s, numerical solutions were still difficult $\Rightarrow$ calibrated/estimated models were not used much

- We will focus on the second-generation literature
- Three influential applications:
 - ① Gourinchas-Parker (Econometrica 2002): Consumption over the Life Cycle
 - ② Blundell-Pistaferri-Preston (AER 2008): Consumption Inequality and Partial Insurance
 - ③ Kaplan-Violante (Econometrica 2014): A Model of the Consumption Response to Fiscal Stimulus Payments
- I use my notation and will make some simplifications when discussing these papers

- Q: How do consumption-savings dynamics evolve over the life cycle?
 - ▶ Does buffer-stock savings behavior or the permanent-income hypothesis provide a better fit of the data? Does it depend on age?
- Method: estimate a structural life-cycle model with idiosyncratic income risk
 - ▶ Estimate income processes in PSID data
 - ▶ Construct consumption-income age profiles using CEX data
 - ▶ Estimate preference parameters to match profiles
 - ▶ Decompose savings behavior over the life cycle using estimated model

- Life-cycle version of the model we have described:

$$\begin{aligned} \max_{C_t, A_{t+1}} \quad & E_0 \sum_{t=0}^T \beta^t \frac{C_t^{1-\sigma}}{1-\sigma} \\ \text{s.t.} \quad & C_t + A_{t+1} = Y_t + RA_t \\ & A_{t+1} \geq 0 \\ & C_t \geq 0 \end{aligned}$$

- For $0 \leq t \leq N$, households are working and income process is

$$\begin{aligned} Y_t &= P_t e^{\epsilon_t} \\ P_t &= P_{t-1} G_t e^{\nu_t} \end{aligned}$$

note that G_t has a t subscript

- For $N + 1 \leq t \leq T$, households are retired and Y_t is deterministic

- $R = 1.034$ is taken from the average return on safe bond assets
- For other parameters, use micro data
 - ▶ PSID (panel with rich info on income but small sample)
 - ▶ CEX (rich consumption data and larger sample, but weak panel dimension)
- Income process:
 - ▶ Estimate age-income profile G_t by regressing income on age dummies with various controls
 - ▶ Estimate permanent-transitory income shocks from income growth residuals
- Consumption-income profiles
 - ▶ Estimate consumption-income age profiles in the data
 - ▶ Set β and σ so that the model mimics this profile as closely as possible
- GP do this procedure for 16 different educational-occupational groups, but here we only focus on the results for the total population

Gourinchas-Parker: Estimated consumption-income profile

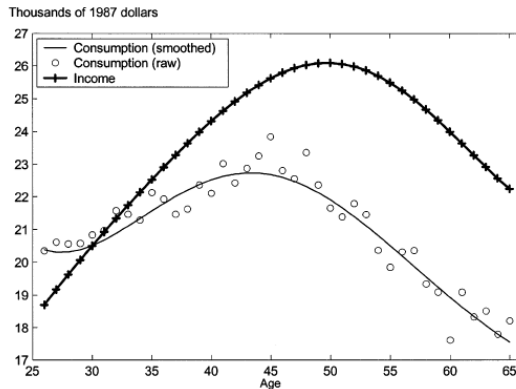


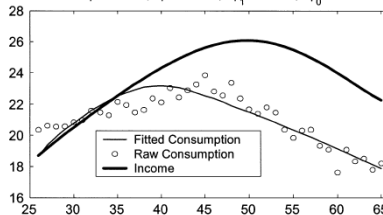
FIGURE 2.—Household consumption and income over the life cycle.

- The hump shape in consumption directly rejects the hypothesis that working-age households have full insurance - why?

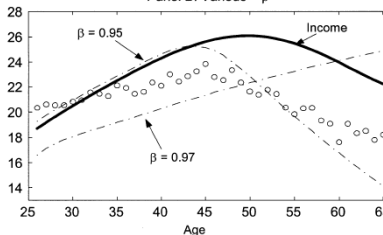
Gourinchas-Parker: Matching model to estimated consumption-income profile

Thousands of 1987 dollars

Panel A: Baseline Estimation
 $\beta = 0.960$, $\rho = 0.514$, $\gamma_1 = 0.071$, $\gamma_0 = 0.001$

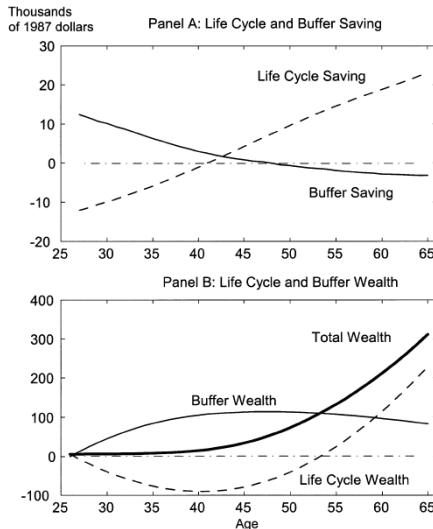


Panel B: Various β



- Matching the consumption-income profile pins down β and σ

Gourinchas-Parker: Results

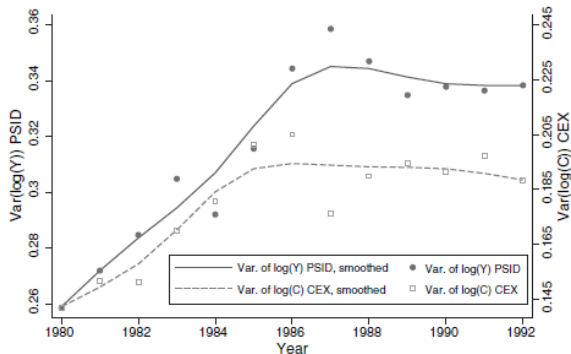


- By turning income risk on and off in the model, GP decompose household savings behavior into its *life-cycle* and *precautionary* components

- Young households are buffer-stock savers, while old households are similar to PIH savers
- Young households: start with no buffer and retirement is far away $\Rightarrow$ saving primarily reflects precautionary behavior
- Middle-aged households: already have a buffer and retirement is near $\Rightarrow$ saving primarily reflects consumption smoothing around retirement

- An early paper estimating a consumption-savings model to match key moments in microdata
- A large literature has followed
- Two related recent papers:
 - ▶ Boar (ReStud 2021): estimates an OLG version of the model using PSID data
 - ★ Finds that a significant part of household buffers is explained by children's income risk
 - ▶ Druedahl and Martinello (ReStat 2020): use Swedish register data to estimate consumption responses to exogenous bequest shocks
 - ★ They find that households quickly consume these bequests $\Rightarrow$ suggests low β
 - ★ Still, these households typically have sizable wealth holdings
 - ★ Matching these moments suggests that risk aversion is much higher than previously estimated

Blundell-Pistaferri-Preston (AER 2008): The question



- In the 1980s, US income inequality grew a lot
- Consumption inequality also grew, but at a lower pace, and the gap between consumption and income inequality increased significantly in the late 1980s/early 1990s
- What can explain this divergence?

- Buffer-stock savings theory teaches us that consumption responds very differently to permanent versus transitory income shocks
- With micro panel data on consumption and income, we can estimate
 - ① The evolution of permanent income risk/inequality
 - ② The evolution of transitory income risk/inequality
 - ③ The evolution of the pass-through of permanent and transitory income to consumption (the extent of insurance)
- For the third part, BPP suggest estimating the effect using a linear consumption rule
 - ▶ People sometimes call this a “semi-structural” approach; here, “semi” mostly means “linear”
- The paper makes two contributions
 - ① Show how to impute consumption in PSID (panel data on income) using food Engel curves
 - ② Use consumption-augmented PSID to estimate the evolution of permanent/transitory income risk and pass-through to consumption
- I focus on the second contribution here

- Suppose we have panel data on log consumption and log income $\{c_{it}, y_{it}\}$
- First, take out deterministic component by regressing c_{it}, y_{it} on various demographic variables
- Second, assume residualized income evolves according to this process (the paper uses a slightly more general process, but it does not affect the point here)

$$y_{it} = p_{it} + \epsilon_{it}$$

$$p_{it} = p_{it-1} + \nu_{it}$$

- Third, assume consumption responds to shocks according to

$$\Delta c_{it} = \phi_t \nu_{it} + \psi_t \epsilon_{it} + \xi_{it}$$

- ▶ Note 1: the treatment effect is time-varying
- ▶ Note 2: this consumption rule is postulated, not derived

- With the postulated consumption rule, we have

$$Var(\Delta c_{it}) = \phi_t^2 Var(\nu_{it}) + \psi_t^2 Var(\epsilon_{it}) + Var(\xi_{it})$$

Assuming that the error term is stationary, consumption growth inequality may grow because

- ① Income variances $Var(\nu_{it})$, $Var(\epsilon_{it})$ increase, or
 - ② Pass-through parameters ϕ_t , ψ_t increase
- Goal: estimate the time series of $Var(\nu_{it})$, $Var(\epsilon_{it})$, ϕ_t , ψ_t
- Method: Estimation by method of moments (we need at least four moments)

- Income process implies

$$\text{Var}(\Delta y_{it}) = \text{Var}(\nu_{it}) + 2\text{Var}(\epsilon_{it})$$

$$\text{Cov}(\Delta y_{it}, \Delta y_{it-1}) = -\text{Var}(\epsilon_{it})$$

- $\Rightarrow$ A rise in income growth inequality without a corresponding decline in autocovariance means that permanent, and not transitory, income variance has increased
- We also have that

$$\begin{aligned}\text{Cov}(\Delta c_{it}, \Delta y_{it}) &= \text{Cov}(\phi_t \nu_{it} + \psi_t \epsilon_{it} + \xi_{it}, \nu_{it} + \epsilon_{it} - \epsilon_{it-1}) \\ &= \phi_t \text{Var}(\nu_{it}) + \psi_t \text{Var}(\epsilon_{it})\end{aligned}$$

- We can then use changes in consumption growth variance:

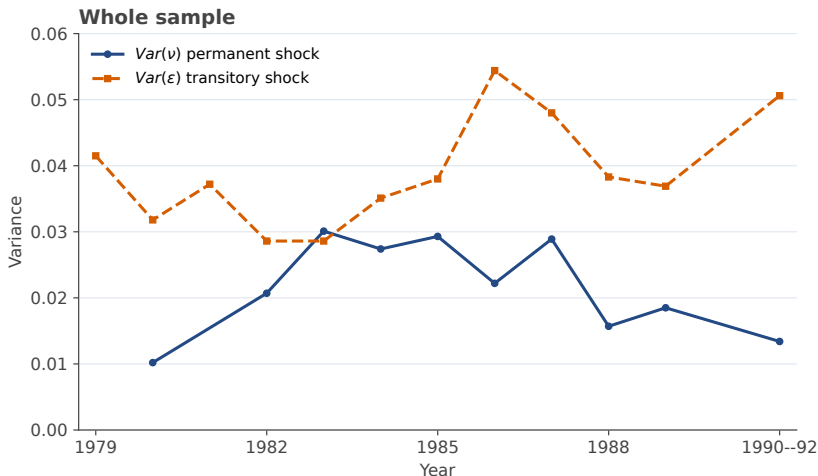
$$\begin{aligned}\Delta \text{Var}(\Delta c_{it}) &= \text{Var}(\nu_{it})\Delta\phi_t^2 + \phi_{t-1}^2\Delta\text{Var}(\nu_{it}) \\ &\quad + \text{Var}(\epsilon_{it})\Delta\psi_t^2 + \psi_t^2\Delta\text{Var}(\epsilon_{it})\end{aligned}$$

to estimate ϕ_t, ψ_t given the income variances

BPP: Pass-through coefficients, full sample

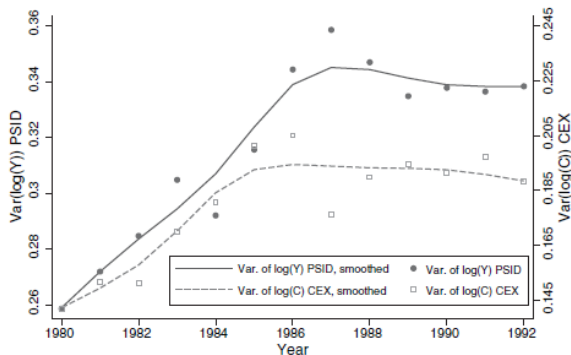
ϕ	0.6423	0.9439	0.4194	0.7928	0.6889
(Partial insurance perm. shock)	(0.0945)	(0.1783)	(0.0924)	(0.1848)	(0.2393)
ψ	0.0533	0.0768	0.0273	0.0675	-0.0381
(Partial insurance trans. shock)	(0.0435)	(0.0602)	(0.0550)	(0.0705)	(0.0737)
<i>p</i> -value test of equal ϕ	23%	99%	8%	81%	18%
<i>p</i> -value test of equal ψ	75%	33%	29%	76%	4%

BPP: Permanent and transitory income variance, full sample



Note: Own illustration based on tables in BPP. Permanent-shock endpoints are reported for 1979–81 and 1990–92; transitory-shock endpoint is reported for 1990–92.

BPP: Back to the question



- In the early 1980s, US consumption inequality tracked income inequality because income inequality growth was driven by permanent-income inequality growth
- The series diverged from the mid-1980s onward, as income inequality growth stemmed from transitory inequality growth, against which households are much more insured

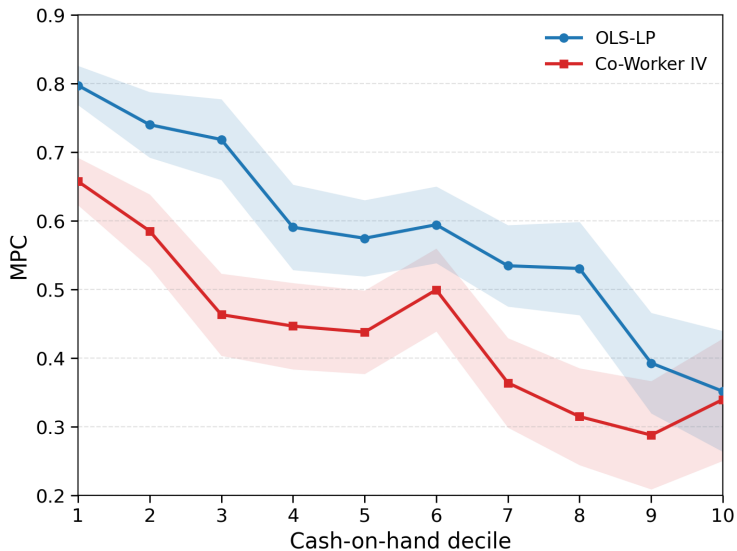
- Why do we care?
 - ▶ Important to understand the nature of income inequality in order to assess welfare implications
 - ▶ An increase in income inequality driven by transitory-income shocks is much less severe than one driven by permanent-income shocks
- See Heathcote-Storesletten-Violante (RED 2023) for documentation of similar facts for later decades

- Great virtue of BPP: robust to i.i.d. measurement error in income
- Big problem of BPP: postulated consumption rule implies random-walk w.r.t. all shocks; not consistent with buffer-stock savings model
- A theoretically consistent approach must allow for state-dependence, e.g.,

$$\Delta c_{it}(m_{it}) = \phi_t(m_{it})\nu_{it} + \psi_t(m_{it})\epsilon_{it} + \xi_{it}$$

but in survey data, there is not enough power to estimate state-dependent parameters

- Nowadays, we have access to large-scale administrative data with minimal measurement error in income
- Carlsson-D'Amico-Oberg-Skans-Walentin (2026): derive an optimal estimator for state-dependent pass-through coefficients under no measurement error in income, maintaining the assumptions of the canonical buffer-stock savings model



- Natural experiments suggest that the within-quarter *aggregate* MPC out of transitory income shocks is quite large
- In 2001 and 2007-2008, the US government provided cash transfers to households as part of stimulus programs, with random timing of the transfer
- Johnson-Parker-Souleles (AER 2006) and Parker-Souleles-Johnson-McClelland (AER 2011) exploit random timing to estimate MPCs of approximately 0.25
- As we saw above, this seems corroborated when applying BPP to high-quality admin data

- We know from buffer-stock theory that individual MPC can be large among wealth-poor households
- However, households with little net assets account for a small share of total consumption $\Rightarrow$ cannot explain why aggregate MPC is large
- How to make sense of this?
- Aggregate MPC is a key moment for understanding business cycles and the macroeconomic effect of stabilization policies

- Idea: even though many households have substantial wealth holdings, much of this wealth is *illiquid*
 - ▶ Housing wealth, retirement accounts...
- It is primarily access to liquid wealth that should determine households' ability to insure against transitory income shocks
- Introduce the concept of “wealthy hand-to-mouth” (wealthy HtM) households: households with a lot of assets but little liquid assets
- Two contributions:
 - ① Document that up to 1/4 of US households can be classified as wealthy HtM households
 - ② Extend buffer-stock savings model to have both liquid and illiquid savings
 - ★ Calibrate to match documented evidence
 - ★ Assess whether model can explain quasi-experimental evidence on aggregate MPC

Kaplan-Violante: Descriptive evidence on asset holdings

	Median (\$2001)	Mean (\$2001)	Fraction Positive	Return (%)
Earnings plus benefits (age 22–59)	41,000	52,745	–	–
Net worth	62,442	150,411	0.90	1.7
Net liquid wealth	2,629	31,001	0.77	–1.5
Cash, checking, saving, MM accounts	2,858	12,642	0.92	–2.2
Directly held MF, stocks, bonds, T-Bills	0	19,920	0.29	1.7
Revolving credit card debt	0	1,575	0.41	–
Net illiquid wealth	54,600	119,409	0.93	2.3
Housing net of mortgages	31,000	72,592	0.68	2.0
Retirement accounts	950	34,455	0.53	3.5
Life insurance	0	7,740	0.27	0.1
Certificates of deposit	0	3,807	0.14	0.9
Saving bonds	0	815	0.17	0.1

^a Authors' calculations based on the 2001 Survey of Consumer Finances (SCF). The return reported in the last column is the real after-tax risk-adjusted return. MM: money market; MF: mutual funds. See Appendix B.1 for additional details.

- Define HtM household as a household with liquid assets $< 1/2$ of monthly earnings
- Define wealthy HtM as HtM households with net illiquid assets $> x$ dollars ($x = 0, 1000, 3000$)
- Depending on x and the definition of illiquid assets, you find that wealthy HtM households make up 7 – 24% of the population

- A simplified version of KV's life-cycle model

$$\begin{aligned} \max_{C_t, A_{t+1}} \quad & E_0 \sum_{t=0}^T \beta^t U(c_t) \\ \text{s.t.} \quad & C_t + A_{t+1}^{\text{liq}} + A_{t+1}^{\text{ill}} = Y_t + R^{\text{liq}} A_t^{\text{liq}} + R^{\text{ill}} A_t^{\text{ill}} - AC(A_{t+1}^{\text{ill}}, A_t^{\text{ill}}) \\ & AC(A_{t+1}^{\text{ill}}, A_t^{\text{ill}}) = \begin{cases} 0 & \text{if } A_{t+1}^{\text{ill}} = R A_t^{\text{ill}} \\ \kappa & \text{if } A_{t+1}^{\text{ill}} \neq R A_t^{\text{ill}} \end{cases} \\ & Y_t = P_t e^{\epsilon_t} \\ & P_t = P_{t-1} G e^{\nu_t} \\ & A_{t+1}^{\text{ill}}, C_t \geq 0 \\ & A_{t+1}^{\text{liq}} \geq -\bar{A} \end{aligned}$$

- Their full model also includes
 - ▶ Utility flow from illiquid assets (think housing)
 - ▶ Different interest rate on borrowing compared to saving
 - ▶ Taxes and transfers
 - ▶ Richer income process
- When taking model to the data, the key difficulty lies in calibrating κ

Kaplan-Violante: Behavior of a wealthy HtM household

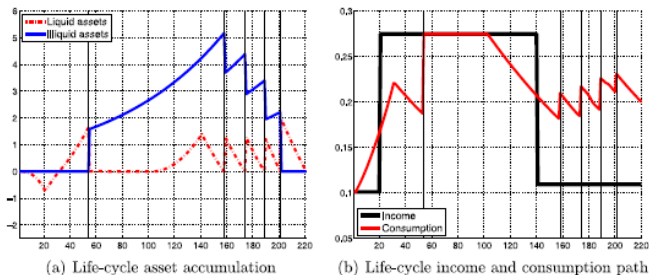
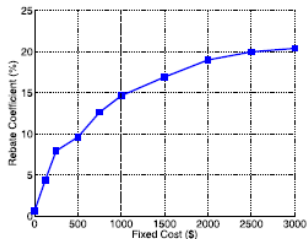


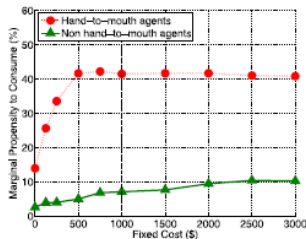
FIGURE 2.—Example of life-cycle of a wealthy hand-to-mouth agent in the model.

- A model period is a quarter; period 0 is age 20
- Since $R^{\text{ill}} > R^{\text{liq}}$, saving in illiquid assets maximizes lifetime consumption
- Fixed cost κ means that you want to make as few transactions in illiquid assets as possible $\Rightarrow$ illiquid assets provide poor insurance
- If the return difference is large enough, households put all cash on hand into the illiquid account when investing, becoming liquidity constrained thereafter

Kaplan-Violante: Implications for the aggregate MPC



(a) Rebate coefficient



(b) Average marginal prop. to consume

FIGURE 5.—Rebate coefficient and marginal propensity to consume, by transaction cost.

- Matching average returns on liquid and illiquid savings, KV finds that even with modest fixed cost, the model can explain high aggregate MPC
- KV also show that this model can match cross-sectional evidence on the distribution of MPC across income and wealth

- In conventional representative-agent macro models, MPC is low, and fluctuations in income therefore play little role in fluctuations in aggregate demand
 - ▶ This is counterintuitive and at odds with micro-level evidence
 - ▶ KV provides a theory for why income fluctuations may matter
 - ▶ Important for the development of heterogeneous-agent New Keynesian (HANK) models (more about this later...)
- An alternative theory to explain high aggregate MPC: some households are simply not very optimizing and consume directly out of what they receive, independently of their financial situation
 - ▶ See, e.g., Campbell-Mankiw (NBER Annual, 1989)
 - ▶ Fagereng-Holm-Natvik (AEJ Macro 2021): using Norwegian register data on consumption responses to lottery winnings, they show that aggregate MPC declines smoothly over time
 - ▶ Auclert-Rognlie-Straub (JPE 2025): an alternative behavioral theory has a hard time explaining the time pattern in MPC, but KV's two-asset model can
- Recent exploratory approaches to understanding heterogeneity in MPCs
 - ▶ Aguiar-Bils-Boar (REStud 2025), Colarieti-Mei-Stantcheva (2025), Carlsson-D'Amico-Öberg-Skans-Walentin (2026b)

Summing up

- Buffer-stock savings model provides a powerful framework for quantitative and empirical research of consumption-savings dynamics
- There is a large literature - we have only glanced at some applications today
- Accompanied by a large literature on estimating household earnings dynamics
- Research in this area operates at the intersection of theory and microdata, often using both structural and reduced-form approaches
- So far, our investigation has been focused on microeconomics: how households behave taking prices, shocks and policy as given
- Next lecture: how do incomplete-market households interact with the aggregate economy?
 - ▶ We need to develop a general equilibrium model